

CSD 380

THE TECHNOLOGY VALUE STREAM

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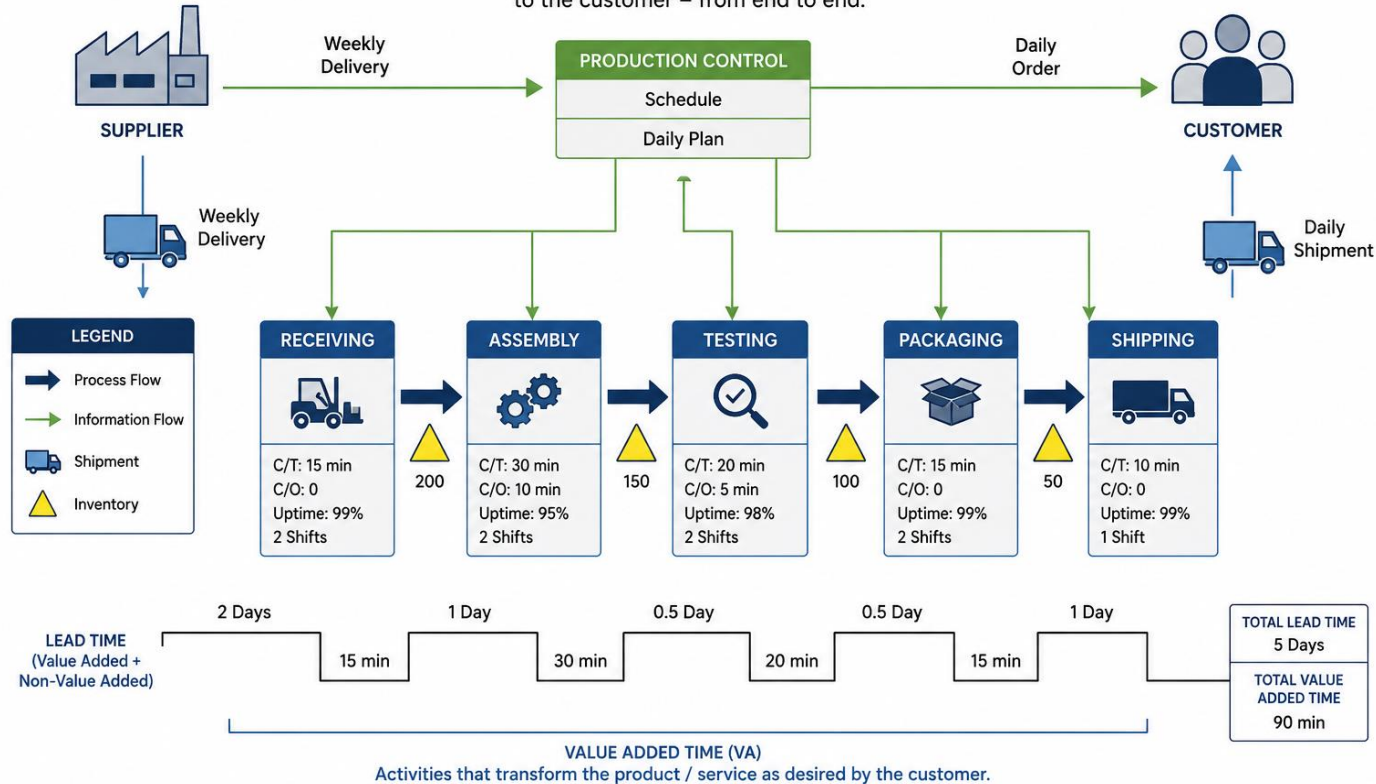
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VALUE STREAM

All the activities required to deliver a product or service to the customer – from end to end.



WHAT IS A VALUE STREAM?

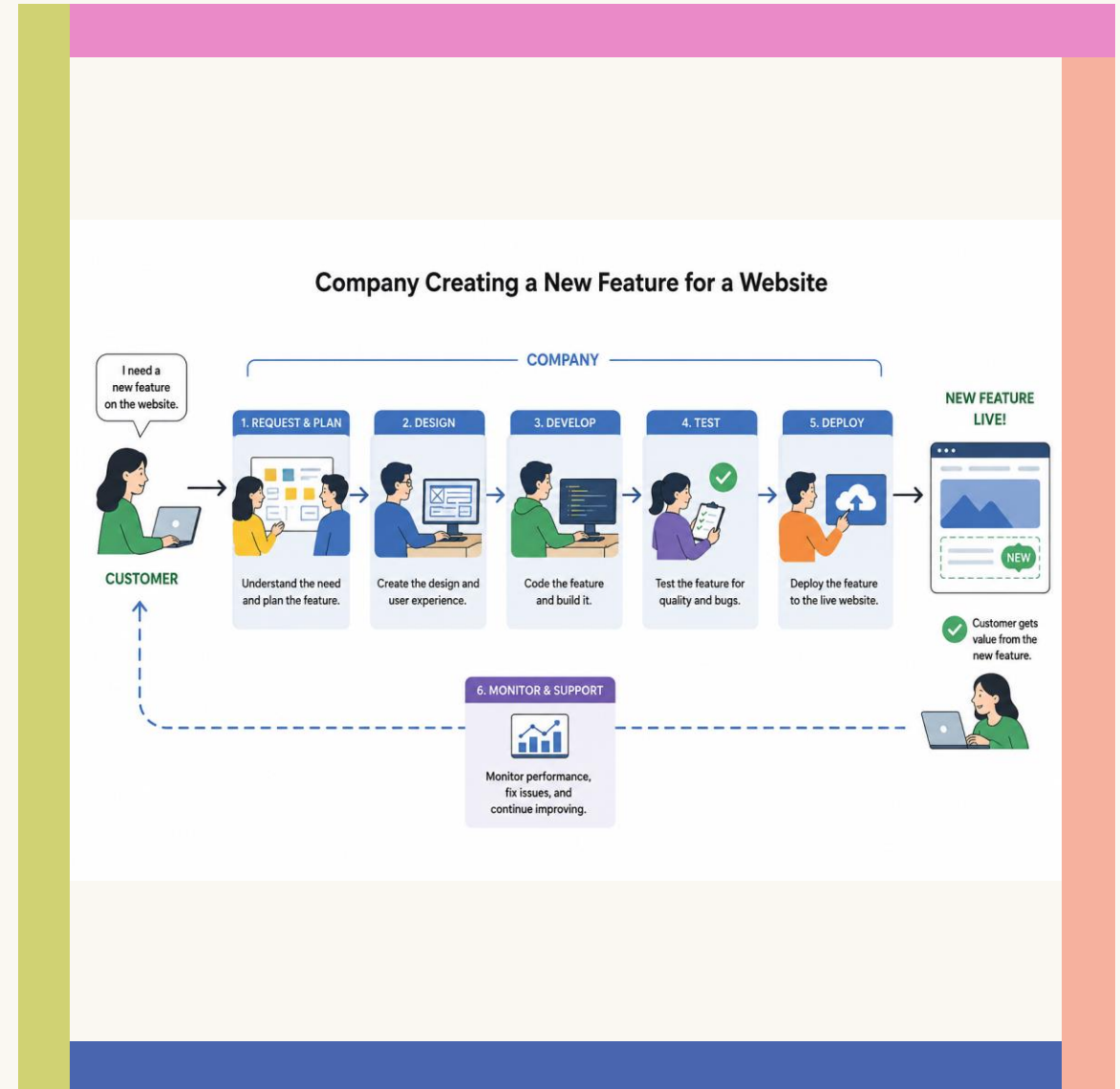
A value stream is the complete sequence of activities needed to deliver something of value to a customer.

THE TECHNOLOGY VALUE STREAM

The "technology value stream" is the series of activities that an organization performs to deliver technology products, services, or software to customers.

When a company creates a new feature for a website, the value stream begins when the feature is requested and ends when it is available for the customer to use.

KIM, G., HUMBLE, J., DEBOIS, P., & WILLIS, J. (2021). *THE DEVOPS HANDBOOK: HOW TO CREATE WORLD-CLASS AGILITY, RELIABILITY, AND SECURITY IN TECHNOLOGY ORGANIZATIONS* (2ND ED.). IT REVOLUTION.



LEAD VS. PROCESSING TIME

- Lead Time is the total amount of time it takes for a task, product, or service to move from the initial customer request to final delivery. Lead time includes both the time work is actively being performed and any waiting time between steps.
- Processing Time is the amount of time spent actively working on the task. It does not include delays, waiting periods, or approvals.
- Lead Time = "Customer creates ticket" through "Customer gets changes"
- Processing Time = "Work/Changes started" through " Work/Changes completed"
- Lead time is what the customer experiences
- Process Time is what the company experiences

DEPLOYMENT LEAD TIMES REQUIRING MONTHS

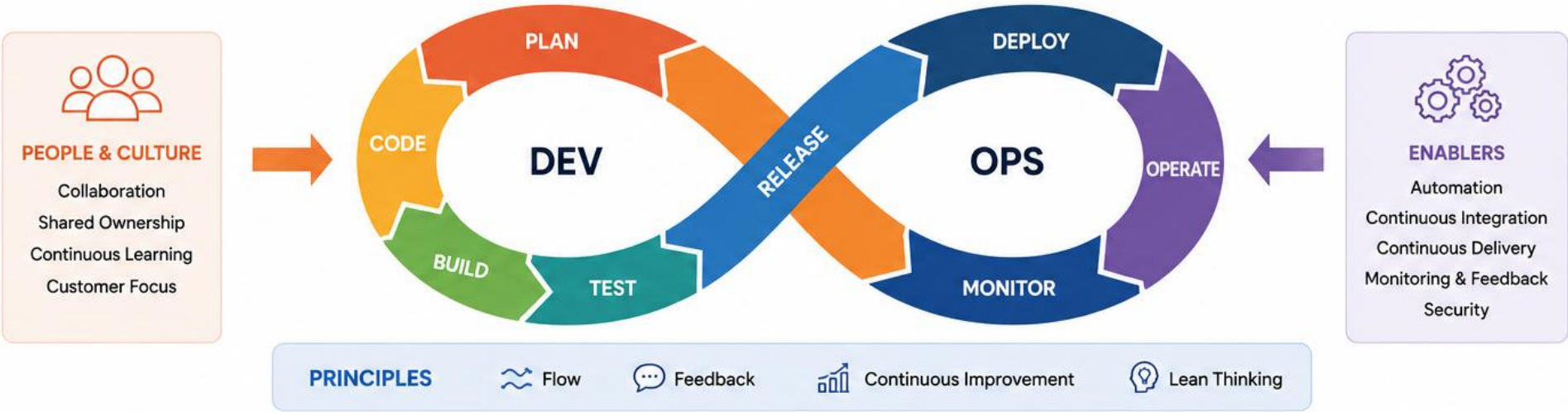
- LEAD TIMES CAN TAKE MONTHS WHEN YOU WORK AT A LARGE OR COMPLEX CORPORATION. THIS IS BECAUSE OF THE DEDICATED APPROVAL PROCESS NEEDED TO BEGIN THE WORK, WHICH CAN CAUSE CONFLICT DOWN THE LINE.
- LONG DEPLOYMENT LEAD TIMES FORCE TEAMS TO COMBINE SUBSTANTIAL AMOUNTS OF CODE AT ONCE, WHICH INCREASES THE LIKELIHOOD OF ERRORS, DELAYS, AND EMERGENCY TROUBLESHOOTING, MAKING SOFTWARE DELIVERY SLOWER AND LESS RELIABLE.
- DEVOPS IS IMPORTANT BECAUSE DEVOPS PRACTICES ENCOURAGE SMALL, FREQUENT DEPLOYMENTS. THIS REDUCES THESE PROBLEMS AND ALLOWS TEAMS TO FIND AND FIX ISSUES MUCH FASTER.

DEPLOYMENT LEAD TIMES OF MINUTES

- To avoid lead times requiring months we use lead times of minutes by using DevOps.
- DevOps focuses on making small, frequent changes, testing them automatically, and deploying them quickly. This reduces errors, speeds up delivery, and helps teams provide value to customers faster.
- Applications are designed in smaller, independent components so teams can work without interfering with one another. If a problem occurs, it is usually limited to a small area instead of affecting the entire system.

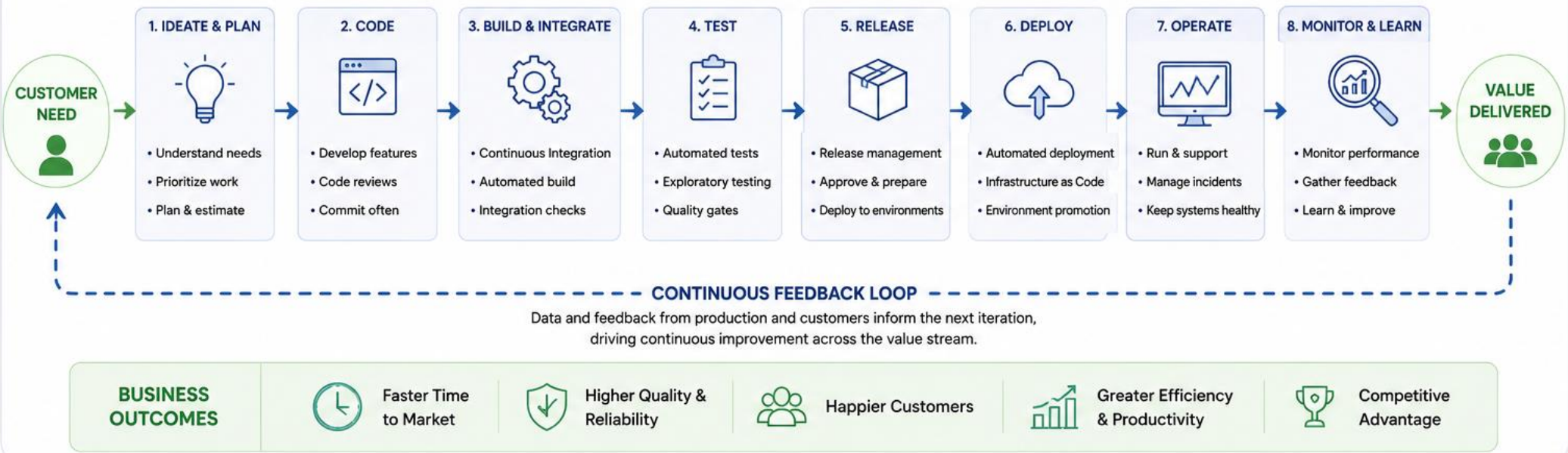
DEVOPS AND THE TECHNOLOGY VALUE STREAM

DevOps is the culture, practices, and automation that enable the Technology Value Stream to deliver value to customers quickly, reliably, and continuously.



THE TECHNOLOGY VALUE STREAM

The end-to-end flow of activities that turns ideas and requirements into value for our customers.



REFERENCES

- KIM, G., HUMBLE, J., DEBOIS, P., & WILLIS, J. (2021). PP 9-10. *THE DEVOPS HANDBOOK: HOW TO CREATE WORLD-CLASS AGILITY, RELIABILITY, AND SECURITY IN TECHNOLOGY ORGANIZATIONS* (2ND ED.). IT REVOLUTION.